Frequency versus Productivity and Why AI Does Not Use the Latter

MANOVA AI

July 20, 2026

To understand why artificial intelligence models do not rely on the linguistic concept of productivity, we must first examine the nature of regular systems. A regular system permits endless, rule-governed operations. The clearest example of a regular, endless system is the set of natural numbers ($N = \{1, 2, 3, 4, \ldots\}$). The constant difference of 1 between adjacent elements makes this sequence an arithmetic progression—a strictly regular set—allowing infinite continuation. In linguistics, infinite generation is understood as full productivity. In natural language, however, even highly productive rules, such as the attachment of the English plural suffix -*s*, encounter irregular formations (*man* → *men*, *child* → *children*). Human language is inherently irregular. This fundamental irregularity makes natural language incompatible with full productivity in the mathematical sense, casting doubt on how useful productivity is for generating language data.

Consequently, computer science sees language as governed by frequency in terms of bounded tokenization and probability distributions over tokens, with token relationships represented through embeddings (combinability specifications). In a system with a defined, bounded size—such as a large language model (LLM)—one can count the exact number of occurrences of every element, which allows the model to process and generate language mathematically.

The Structural Unit: Tokens over Morphemes and Words

In an AI model, text is processed as a continuous linear sequence of characters segmented into tokens. Tokens serve as the fundamental units of the model, and frequency of occurrence is the sole criterion for assigning token status to a character sequence. A token can represent any arbitrary string: individual letters, subword units, whole words, leading spaces, punctuation, coding syntax, or emojis.

Crucially, a token is a purely formal and structural unit—a slice of raw text completely detached from semantic meaning—assigned a specific integer ID number. The AI model does not “see” letters, words, or meanings; it operates exclusively on these ID numbers. The set of all tokens of an AI model represents the model’s vocabulary.

Modern models operate with a fixed, known vocabulary of tokens, typically ranging between 50,000 and 200,000 discrete IDs. Thus, while the vocabulary of a language and the mental lexicon of an individual speaker are open sets that may expand through creativity and productivity, an AI model’s token list is a strictly closed set that cannot be expanded. Furthermore, these tokens are ordered sequentially by frequency of use, with the most frequent tokens receiving the lowest ID numbers (see ChatGPT’s list of tokens at emaggiori.com/chatgpt-all-tokens/ for illustration). It should be noted that other tokenization models also exist.

Settled Systems versus Dynamic Systems

Tokenization turns an AI model into a settled, static system. Every token is defined mathematically by its combinability with all other tokens in the closed vocabulary (list of tokens). These structural relationships are represented as continuous numerical vectors called embeddings. Every token is mapped across exactly the same number of dimensions within a vector space. There is only one vector (derivation) space for all languages, and all derivations take place there. Actually, the only derivation operation that takes place in an AI model is the addition of a token (cf. Merge in syntax).

In this architecture, frequency is not only the sole criterion for creating a token but also the sole criterion for defining its combinability. In short: frequency governs, structures, and generates the entire system.

By contrast, productivity—a cornerstone of linguistic analyses—presupposes an unsettled, dynamic system. In human language, certain patterns are more productive than others, allowing speakers to derive novel items of varying lengths across words, phrases, and sentences. Crucially, linguistic productivity is rarely quantified with the precision that frequency allows. As linguists often note, a highly productive pattern is not necessarily a highly frequent one. For instance, if the productivity criterion is combinability with a foreign element, the attachment of the English plural *-s* to an intrinsically infrequent pattern in a language can turn this unproductive pattern into a productive one immediately. This also demonstrates the subjective nature of productivity evaluations.

The Problem with Semantics in Computation

Establishing productivity requires coupling form with semantics: as a rule, a structural pattern is considered productive only if it conveys a specific, rule-governed meaning. However, integrating semantics introduces computational complexity:

- Form-based analysis relies on discrete, observable structural slices.
- Semantic-based analysis introduces subjective interpretation, context dependency, ambiguity, and abstraction.

While semantics gives linguists additional space for theoretical analyses, it makes automated, exact computational generation harder. Modeling language exclusively through a closed token set and co-occurrence frequencies—without requiring a model to represent intrinsic meaning—makes the system settled and language generation computationally easier.

Derivation driven by open-ended productivity, on the other hand, is inherently difficult to control. If new structures can be appended to an irregular system, that system can never be generated comprehensively. Even widely productive English affixes such as *-s* and *-ness* do not attach universally to every noun or adjective, respectively.

Conclusion

The distinction between frequency and productivity explains a major divergence between disciplines:

Frequency (Computer Science): Computer scientists, using discrete token units within a settled linear system, can generate fluent, coherent language rapidly at scale.

Productivity (Linguistics): Linguists remain focused on describing open, dynamic systems, identifying productive patterns, and analyzing potential versus impossible coinages.

From a mathematical perspective, treating an irregular human language as an open, dynamic system renders automated fluent language generation an unsolvable problem. By reframing language as a closed, settled system driven entirely by statistical frequency, computer science bypassed this complexity. Language generation built on frequency of use provides a working computational model for human communication, whereas an open system built on dynamic productivity and semantics remains too complex and inefficient for language generation.

Finally, while one can ask an AI model questions about linguistic productivity, its answers depend entirely on the training data it has ingested. The model does not internally utilize or understand “productivity”—it merely reproduces statistical patterns found in human texts, lacking any internal mechanism to judge whether its conceptual assertions are objectively right or wrong.

Dr. Stela MANOVA

MANOVA AI / Gauss:AI Global